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Transmission Lines

GATE ECE Electromagnetics · Previous Year Questions with Answer Key

64	39	1987 – 2026	English
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Answer key	at the end

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Marks 1 27 questions

Q1. GATE ECE 2026

Open solution

For a lossless passive two-port network, $|S_{11}|$ and $|S_{21}|$ intersect at -3 dB .

For a lossy passive two-port network, $|S_{11}|$ and $|S_{21}|$ intersect at -4 dB ..

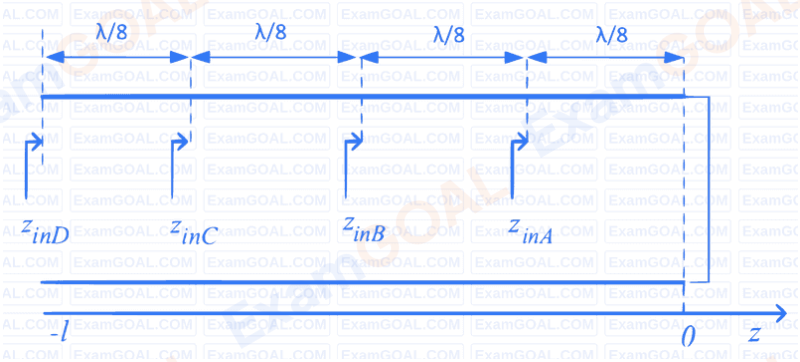
The percentage of power dissipated in the lossy network at the intersection frequency is ____ .

(rounded off to two decimal places)

Q2. GATE ECE 2024

Open solution

Consider a lossless transmission line terminated with a short circuit as shown in the figure below. As one moves towards the generator from the load, the normalized impedances Z_{inA}, Z_{inB}, Z_{inC}, and Z_{inD} (indicated in the figure) are _____.



(A) $Z_{inA} = +1j\Omega$, $Z_{inB} = \infty$, $Z_{inC} = -1j\Omega$, $Z_{inD} = 0$

(B) $Z_{inA} = \infty$, $Z_{inB} = +0.4j\Omega$, $Z_{inC} = 0$, $Z_{inD} = +0.4j\Omega$

(C) $Z_{inA} = -1j\Omega$, $Z_{inB} = 0$, $Z_{inC} = +1j\Omega$, $Z_{inD} = \infty$

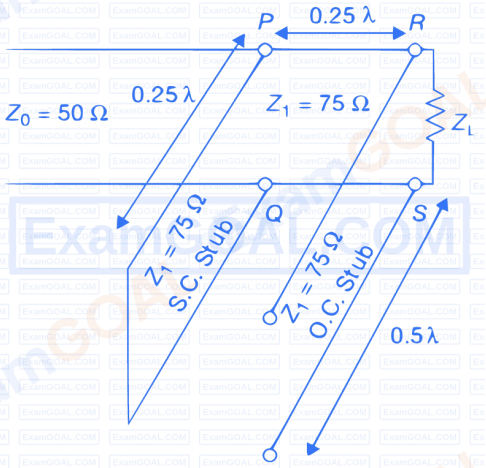
(D) $Z_{inA} = +0.4j\Omega$, $Z_{inB} = \infty$, $Z_{inC} = -0.4j\Omega$, $Z_{inD} = 0$

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Q3. GATE ECE 2021

Open solution

The impedance matching Network shown in figure is to match a lossless line having characteristic impedance $Z_o = 50\Omega$ with a load impedance Z_L. A quarter - Wave line having a characteristic impedance $Z_1 = 75\Omega$ is connected to Z_L. Two stubs having characteristic impedance of 75Ω each are connected to this quarter - wave line. One is a short - circuited (S.C) stub of length 0.25λ connected across PQ and the other one in an open - Circuted (O.C) stub of length 0.5 λ connected across RS.



The impedance matching is achieved when the real part of Z_L is

(A) 33.3Ω (B) 75.0Ω (C) 50.0Ω (D) 112.5Ω

Q4. GATE ECE 2020

Open solution

A transmission line of length $3\lambda/4$ and having a characteristic impedance of 50Ω is terminated with a load of 400Ω. The impedance (rounded off to two decimal places) seen at the input end of the transmission line is ____ Ω.

Q5. GATE ECE 2020

Open solution

The impedances $Z = jX$, for all X in the range $(-\infty, \infty)$, map to the Smith chart as

(A) a circle of radius 1 with centre at $(0, 0)$.
xx

(B) a line passing through the centre of the chart.

(C) a circle of radius 0.5 with centre at $(0.5, 0)$.

(D) a point at the centre of the chart.

Q6. GATE ECE 2017 Set 2

Open solution

A two wire transmission line terminates in a television set. The VSWR measured on the line is 5.8. The percentage of power that is reflected from the television set is

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Q7. GATE ECE 2017 Set 1

Open solution

The voltage of an electromagnetic wave propagating in a coaxial cable with uniform characteristic impedance is $V(l) = e^{-\gamma l + j\omega t}$ Volts, where l is the distance along the length of the cable in metres, $\gamma = (0.1 + j40)\ m^{-1}$ is the complex propagation constant, and $\omega = 2\pi \times 10^9$ rad/s is the angular frequency. The absolute value of the attenuation in the cable in dB/metre is _____.

Q8. GATE ECE 2016 Set 1

Open solution

The propagation constant of a lossy transmission line is (2 + j5) m^{-1} and its characteristic impedance is (50 + j0) Ω at $\omega = 10^6\ rad\ S^{-1}$. The values of the line constants L, C, R, G are, respectively,

(A) $L = 200\,\mu H/m,\ C = 0.1\ \mu F/m,$
$R = 50\ \Omega/m,\ G = 0.02\,S/m,$

(B) $L = 250\,\mu H/m,\ C = 0.1\ \mu F/m,$
$R = 100\ \Omega/m,\ G = 0.04\,S/m,$

(C) $L = 200\,\mu H/m,\ C = 0.2\ \mu F/m,$
$R = 100\ \Omega/m,\ G = 0.02\,S/m,$

(D) $L = 250\,\mu H/m,\ C = 0.2\ \mu F/m,$
$R = 50\ \Omega/m,\ G = 0.04\,S/m,$

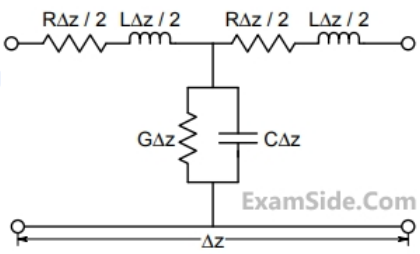
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Q9. GATE ECE 2015 Set 3

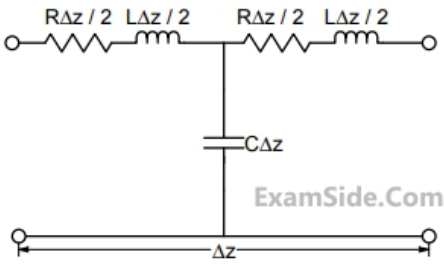
Open solution

A coaxial cable is made of two brass conductors. The spacing between the conductors is filled with Teflon ($\varepsilon_r = 2.1,\ \tan\ \delta = 0$). Which one of the following circuits can represent the lumped element model of a small piece of this cable having length $\Delta\ z$?

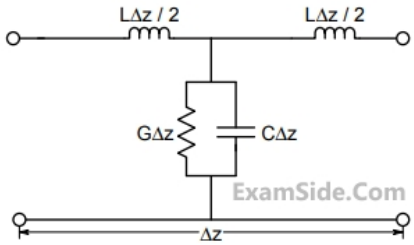
(A)



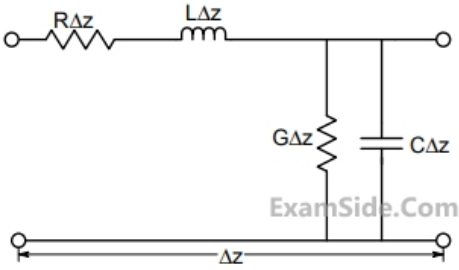
(B)



(C)



(D)

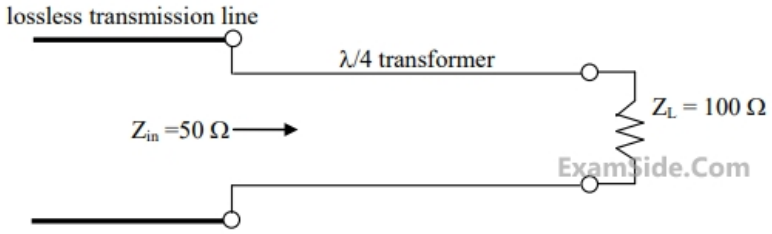


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Q10. GATE ECE 2014 Set 2

Open solution

To maximize power transfer, a lossless transmission line is to be matched to a resistive load impedance via a $\lambda/4$ transformer as shown.

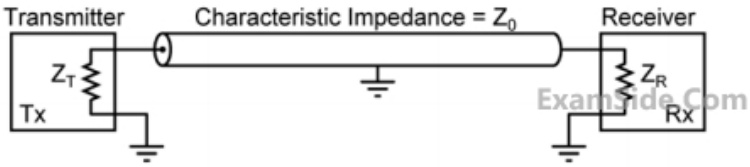


The characteristic impedance (in Ω) of the λ/4 transformer is _____.

Q11. GATE ECE 2014 Set 3

Open solution

In the following figure, the transmitter Tx sends a wideband modulated RF signal via a coaxial cable to the receiver Rx. The output impedance Z_T of Tx, the characteristic impedance Z_0 of the cable and the input impedance Z_R of Rx are all real.



Which one of the following statements is TRUE about the distortion of the received signal due to impedance mismatch?

(A) The signal gets distorted if $Z_R \neq Z_0$, irrespective of the value of Z_T

(B) The signal gets distorted if $Z_T \neq Z_0$, irrespective of the value of Z_R

(C) Signal distortion implies impedance mismatch at both ends: $Z_T \neq Z_0$ and $Z_R \neq Z_0$

(D) Impedance mismatches do NOT result in signal distortion but reduce power transfer efficiency

Q12. GATE ECE 2013

Open solution

The return loss of a device is found to be 20 dB. The voltage standing wave ratio (VSWR) and magnitude of reflection coefficient are respectively.

(A) 1.22 and 0.1 (B) 0.81 and 0.1 (C) - 1.22 and 0.1 (D) 2.44 and 0.2

Q13. GATE ECE 2012

Open solution

A coaxial cable with an inner diameter of 1 mm and outer diameter of 2.4 mm is filled with a dielectric of relative permittivity 10.89. Given $\mu_0 = 4\pi \times 10^{-7}$ H/m, $\varepsilon_0 = \frac{10^{-9}}{36\pi}$ F/m, the characteristic impedance of the cable is

(A) 330 Ω (B) 100 Ω (C) 143.3 Ω (D) 15.9 Ω

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Q14. GATE ECE 2011

A transmission line of characteristic impedance 50 Ω is terminated by a 50 Ω load. When excited by a sinusoidal voltage source at 10 GHz, the phase difference between two points spaced 2 mm apart on the line is found to be $\pi/4$ radians. The phase velocity of the wave along the line is

(A) $0.8 \times 10^8\, m/s$ (B) $1.2 \times 10^8\, m/s$ (C) $1.6 \times 10^8\, m/s$ (D) $3 \times 10^8\, m/s$

Open solution

Q15. GATE ECE 2010

A transmission line has a characteristic impedance of 50 Ω and a resistance of 0.1 Ω/m. If the line is distortionless, the attenuation constant (in Np/m) is

(A) 500 (B) 5 (C) 0.014 (D) 0.002

Open solution

Q16. GATE ECE 2010

If the scattering matrix [S] of a two port network is

$$[S] = \begin{bmatrix} 0.2\angle 0^\circ & 0.9\angle 90^\circ \\ 0.9\angle 90^\circ & 0.1\angle 90^\circ \end{bmatrix}$$

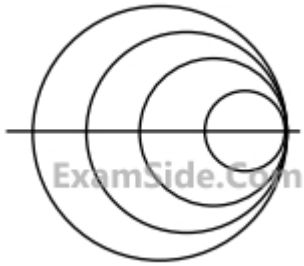
then the network is

(A) lossless and reciprocal
(B) lossless but not reciprocal
(C) not lossless but reciprocal
(D) neither lossless nor reciprocal

Open solution

Q17. GATE ECE 2005

Many circles are drawn in a Smith chart used for transmission line calculations. The circles shown in Fig. represent



(A) unit circles
(B) constant resistance circles.
(C) constant reactance circles.
(D) constant reflection coefficient circles.

Open solution

Q18. GATE ECE 2002

The VSWR can have any value between

(A) 0 and 1 (B) - 1 and + 1 (C) 0 and ∞ (D) 1 and ∞

Open solution

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Q19. GATE ECE 2002

In an impedance Smith chart, a clockwise movement along a constant resistance circle gives rise to

(A) a decrease in the value of reactance
(B) an increase in the value of reactance
(C) no change in the reactance value
(D) no change in the impedance value

Open solution

Q20. GATE ECE 2001

A transmission line is distortionless if

(A) $RL = \frac{1}{GC}$
(B) $RL = GC$
(C) $LG = RC$
(D) $RG = LC$

Open solution

Q21. GATE ECE 2000

The magnitudes of the open-circuit and short-circuit input impedances of a transmission line are 100Ω and 25Ω respectively. The characteristic impedance of the line is

(A) $25\,\Omega$
(B) $50\,\Omega$
(C) $75\,\Omega$
(D) $100\,\Omega$

Open solution

Q22. GATE ECE 1999

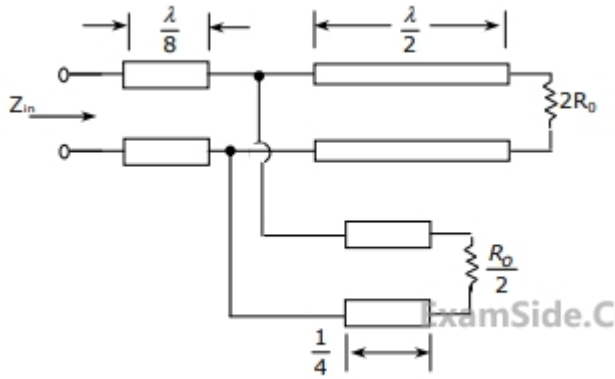
Assuming perfect conductors of a transmission line, pure TEM propagation is NOT possible in

(A) coaxial cable
(B) air-filled cylindrical waveguide
(C) parallel twin-wire line in air
(D) semi-infinite parallel plate waveguide

Open solution

Q23. GATE ECE 1998

All transmission line sections shown in Fig. have characteristic impedance $R_0 + j0$. The input impedance Z_{in} equals



(A) $\frac{2}{3}R_0$
(B) R_0
(C) $\frac{3}{2}R_0$
(D) $2\,R_0$

Open solution

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Q24. GATE ECE 1997

A transmission line of 50Ω characteristic impedance is terminated with a 100 Ω resistance. The minimum impedance measured on the line is equal to

(A) 0 Ω (B) 25 Ω (C) 50 Ω (D) 100 Ω

Open solution

Q25. GATE ECE 1996

A lossless transmission line having 50 Ω characteristic impedance and lenght $\lambda/4$ is short ciruited at one end and connected to an ideal voltage sourec of 1 V at the other end. The current drawn from the voltage source is

(A) 0 (B) 0.02 A

(C) ∞ (D) none of the above

Open solution

Q26. GATE ECE 1996

The capacitance per unit length and the characteristic impedance of a lossless transmission line are C and Z_0 respectively. The velocity of a traveling wave on the transmission line is

(A) Z_0C (B) $\frac{1}{Z_0C}$ (C) $\frac{Z_0}{C}$ (D) $\frac{C}{Z_0}$

Open solution

Q27. GATE ECE 1994

A load impedance (200 + j0) Ω is to be matched to a 50Ω lossless transmission line by using a quarter wave line transformer (QWT). The characteristic impedance of the QWT required is

Open solution

Marks 2 34 questions

Q28. GATE ECE 2026

A complex load (in Ω) is represented as $\Gamma_L = 0.5\angle 30^\circ$ on the Smith chart. A co-axial cable with a characteristic impedance of 50Ω is connected to the load. The new input impedance of the load now moves to a diametrically opposite point on the same Γ circle on the Smith chart.

Which option is the nearest input impedance of the cable connected load (in Ω)?

(A) $20.7 - j5.1$ (B) $17.7 - j11.8$ (C) $97.5 - j65.0$ (D) $97.5 + j65.0$

Open solution

Q29. GATE ECE 2025

A 50Ω lossless transmission line is terminated with a load Z_L of $(50 - j75)\Omega$. If the average incident power on the line is 10 mW , then the average power delivered to the load (in mW , rounded off to one decimal place) is _____ .

Open solution

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Q30. GATE ECE 2024

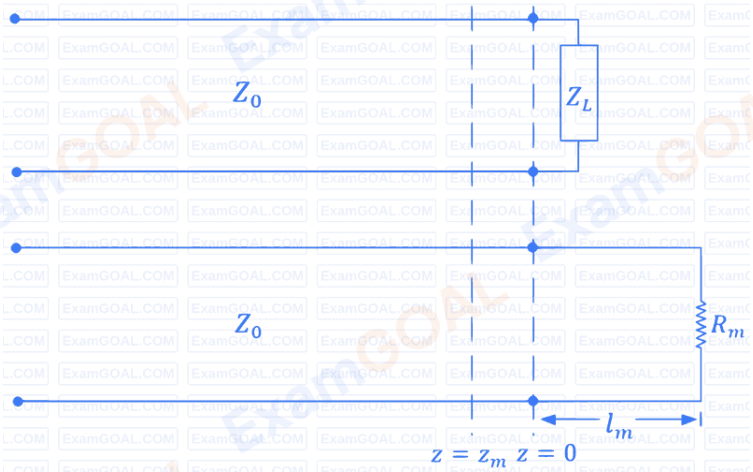
A lossless transmission line with characteristic impedance $Z_0 = 50\Omega$ is terminated with an unknown load. The magnitude of the reflection coefficient is $|\Gamma| = 0.6$. As one moves towards the generator from the load, the maximum value of the input impedance magnitude looking towards the load (in Ω) is _____.

Open solution

Q31. GATE ECE 2023

The standing wave ratio on a 50 Ω lossless transmission line terminated in an unknown load impedance is found to be 2.0. The distance between successive voltage minima is 30 cm and the first minimum is located at 10 cm from the load. Z_L can be replaced by an equivalent length l_m and terminating resistance R_m of the same line. The value of R_m and l_m, respectively, are

Open solution



(A) $R_m = 100\,\Omega, l_m = 20$ cm

(B) $R_m = 25\,\Omega, l_m = 20$ cm

(C) $R_m = 100\,\Omega, l_m = 5$ cm

(D) $R_m = 25\,\Omega, l_m = 5$ cm

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Q32. GATE ECE 2023

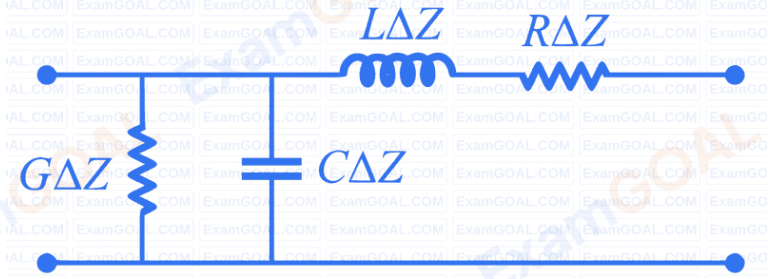
Open solution

The following circuit(s) representing a lumped element equivalent of an infinitesimal sectioin of a transmission line is/are

(A)



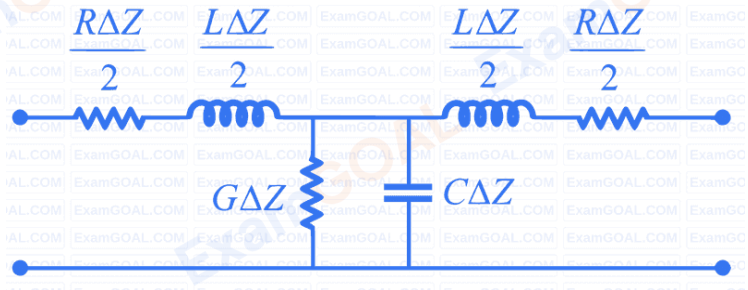
(B)



(C)



(D)

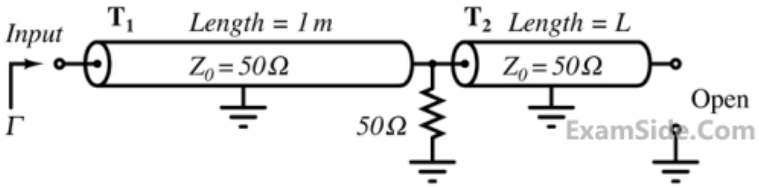


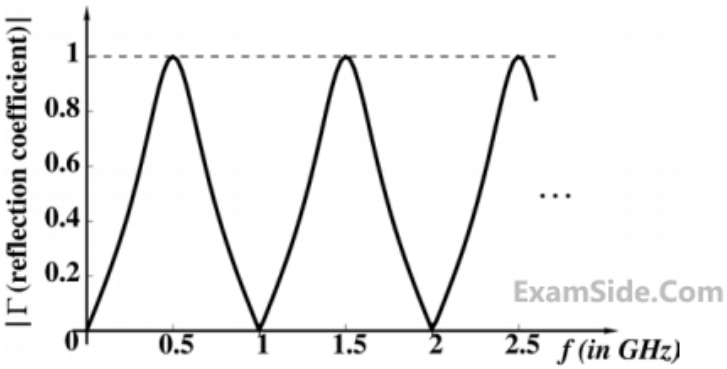
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Q33. GATE ECE 2016 Set 2

Open solution

A microwave circuit consisting of lossless transmission lines T_1 and T_2 is shown in the figure. The plot shows the magnitude of the input reflection coefficient Γ as a function of frequency f. The phase velocity of the signal in the transmission lines is 2×10^8 m/s

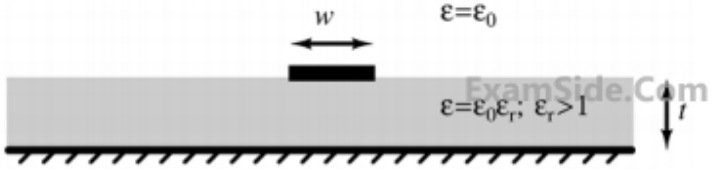




Q34. GATE ECE 2016 Set 2

Open solution

A lossless microstrip transmission line consists of a trace of width w. It is drawn over a practically infinite ground plane and is separated by a dielectric slab of thickness t and relative permittivity $\varepsilon_r > 1$. The inductance per unit length and the characteristic impedance of this line are L and Z_o respectively.



Which one of the following inequalities is always satisfied?

(A) $Z_o > \sqrt{\frac{Lt}{\varepsilon_o \varepsilon_r w}}$

(B) $Z_o < \sqrt{\frac{Lt}{\varepsilon_o \varepsilon_r w}}$

(C) $Z_o > \sqrt{\frac{Lw}{\varepsilon_o \varepsilon_r t}}$

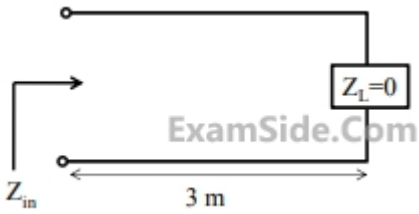
(D) $Z_o < \sqrt{\frac{Lw}{\varepsilon_o \varepsilon_r t}}$

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Q35. GATE ECE 2015 Set 3

Consider the 3 m long lossless air-filled transmission line shown in the figure. It has a characteristic impedance of $120\pi\,\Omega$, is terminated by a short circuit, and is excited with a frequency of 37.5 MHz. What is the nature of the input impedance (Z_{in})?

Open solution

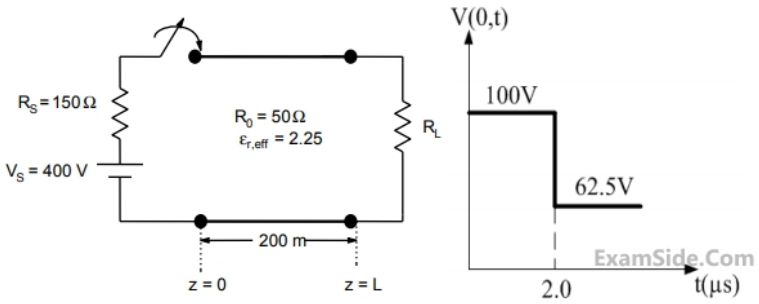


(A) Open (B) Short (C) Inductive (D) Capacitive

Q36. GATE ECE 2015 Set 3

A 200 m long transmission line having parameters shown in the figure is terminated into a load R_L. The line is connected to a 400 V source having source resistance R_S through a switch, which is closed at t = 0. The transient response of the circuit at the input of the line (z = 0) is also drawn in the figure. The value of R_L (in Ω) is _____.

Open solution



Q37. GATE ECE 2014 Set 1

For a parallel plate transmission line, let v be the speed of propagation and Z be the characteristic impedance. Neglecting fringe effects, a redution of the spacing between the plates by a factor of two results in

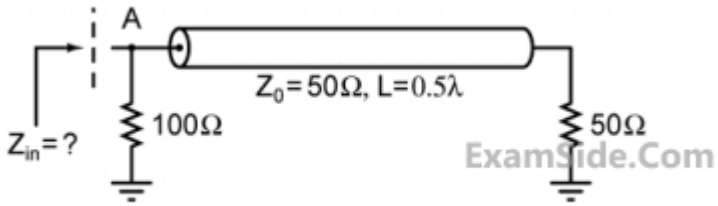
Open solution

(A) halving of v and no change in Z
(B) no changes in v and halving of Z
(C) no change in both v and Z
(D) halving of both v and Z

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Q38. GATE ECE 2014 Set 2

In the transmission line shown, the impedance Z_{in} (in ohms) between node A and the ground is



Open solution

Q39. GATE ECE 2012

A transmission line with a characteristic impedance of 100 Ω is used to match a 50 Ω section to a 200 Ω section. If the matching is to be done both at 429 MHz and 1 GHz, the tength of the transmission line can be approximately

(A) 82.5 cm (B) 1.05 m (C) 1.58 m (D) 1.75 m

Open solution

Q40. GATE ECE 2011

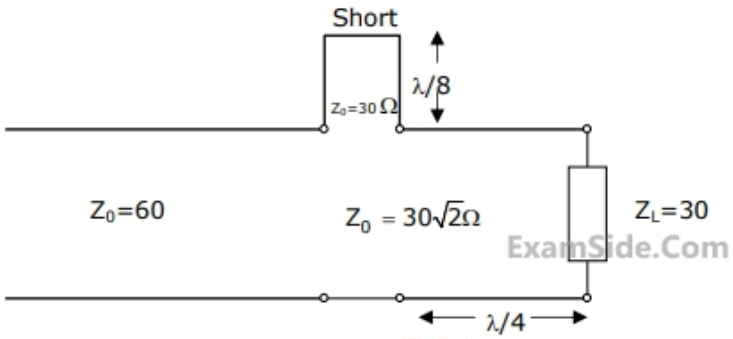
A transmission line of characteristic impedance 50 Ω is terminated in a load impedance Z_L. The VSWR of the line is measured as 5 and the first of the voltage naxima in the line is observed at a distance of $\lambda/4$ from the load. The value of Z_L is

(A) 10 Ω (B) 250 Ω
(C) (19.23 + j 46.15) Ω (D) (19.23 - j 46.15)Ω

Open solution

Q41. GATE ECE 2010

In the circuit shown, all the transmission line sections are lossless. The Voltage Standing Wave Ratio (VSWR) on the 60 Ω line is



(A) 1.00 (B) 1.64 (C) 2.50 (D) 3.00

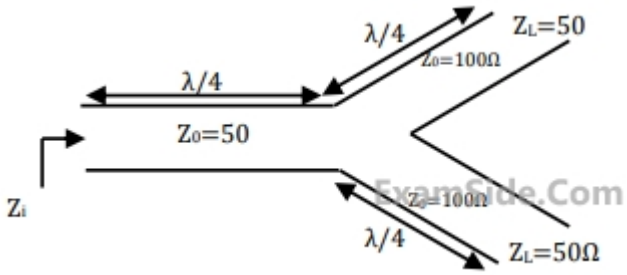
Open solution

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Q42. GATE ECE 2009

A transmission line terminates in two branches, each of length $\lambda/4$, as shown.The branches are terminated by 50 Ω loads. The lines are lossless and have the characteristic impedances shown. Determine the impedance Z_i as seen by the source.

Open solution



(A) 200 Ω (B) 100 Ω (C) 50 Ω (D) 25 Ω

Q43. GATE ECE 2008

One end of loss-less transmission line having the characteristic impedance of 75 Ω and length of 1 cm is short-ciruited. At 3 GHz, the input impedance at the other end of the transmission line is

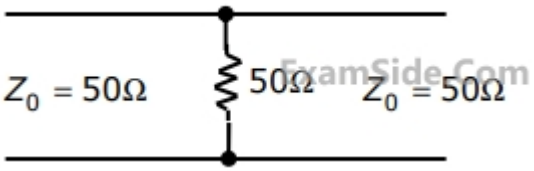
Open solution

(A) 0 (B) Resistive (C) Capacitive (D) Inductive

Q44. GATE ECE 2007

A load of 50 Ω is connected in shunt in a 2-wire transmission line of Z_0 = 50 Ω as shown in the Fig. The 2-port scattering parameter matrix (S-matrix) of the shunt element is

Open solution



(A) $\begin{bmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$

(B) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(C) $\begin{bmatrix} -\frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} \end{bmatrix}$

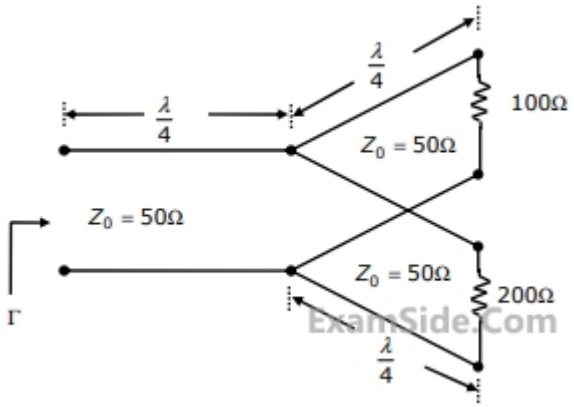
(D) $\begin{bmatrix} \frac{1}{4} & -\frac{3}{4} \\ -\frac{3}{4} & -\frac{1}{4} \end{bmatrix}$

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Q45. GATE ECE 2007

The parallel branches of a 2-wire transmission line are terminated in 100 Ω and 200 Ω resistors as shown in the Fig. The characteristic impedance of the line is Z_0 = 50 Ω and each section has a length of $\lambda/4$. The voltage reflection coefficient Γ at the input is:

Open solution



(A) $-j\frac{7}{5}$ (B) $-\frac{5}{7}$ (C) $j\frac{5}{7}$ (D) $\frac{5}{7}$

Q46. GATE ECE 2005

Characteristic impedance of a transmission line is 50Ω. Input impedance of the open-circuited line is $Z_{oc} = 100 + j\ 150\Omega$. When the transmission line is short-circuited the value of the input impedance will be

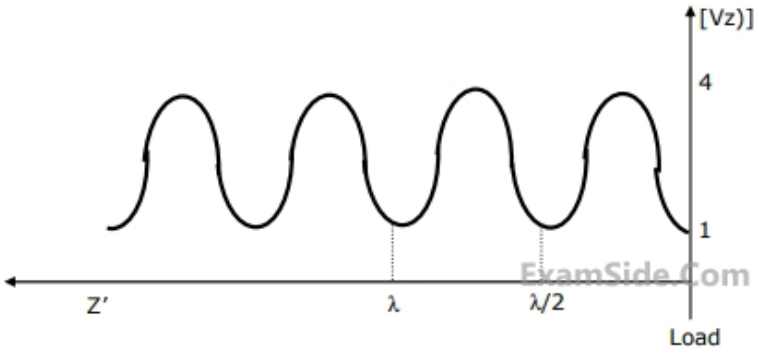
Open solution

(A) 50 Ω (B) 100 + j 150 Ω
(C) 7.69 + j 11.54 Ω (D) 7.69 - j 11.54 Ω

Q47. GATE ECE 2005

Voltage standing wave pattern in a lossless transmission line with characteristic impedance 50 Ω and a resistive load is shown in Fig.

Open solution



The value of the load resistance is

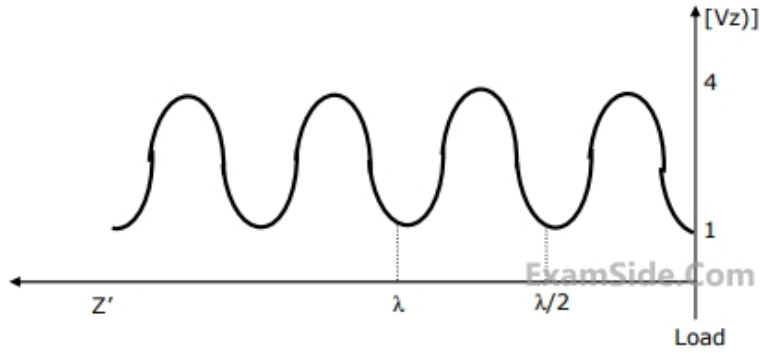
(A) 50 Ω (B) 200 Ω (C) 12.5 Ω (D) 0 Ω

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Q48. GATE ECE 2005

Open solution

Voltage standing wave pattern in a lossless transmission line with characteristic impedance 50 Ω and a resistive load is shown in Fig.



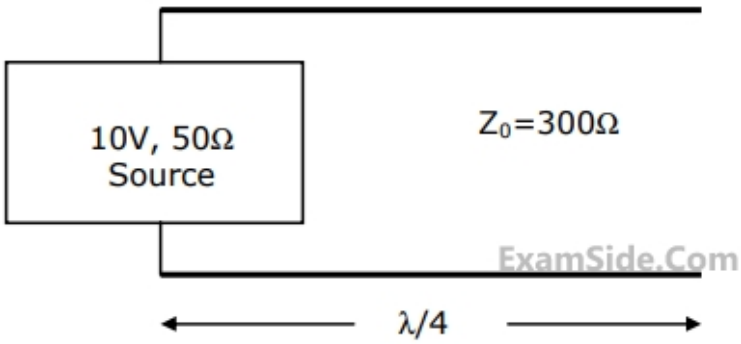
The reflection coefficient is given by

(A) - 0.6 (B) - 1 (C) 0.6 (D) 0

Q49. GATE ECE 2004

Open solution

Consider a 300Ω, quarter-wave long (at 1 GHz) transmission line as shown in Fig. It is connected to a 10V, 50Ω sources at one end and is left open circuited at the other end. The magnitude of the voltage at the open circuit end of the line is



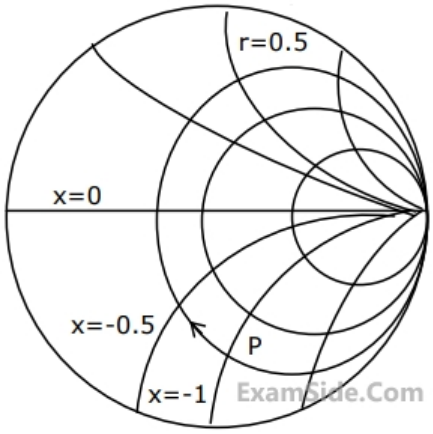
(A) 10 V (B) 5 V (C) 60 V (D) 60/7 V

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Q50. GATE ECE 2004

Open solution

Consider an impedance Z = R + jX marked with point P in an impedance Smith chart as shown in Fig. The movement from point P along a constant resistance circle in the clockwise direction by an angle 45° is equivalent to



(A) adding an inductance in series with Z
(B) adding a capacitance in series with Z
(C) adding an inductance in shunt across Z
(D) adding a capacitance in shunt across Z

Q51. GATE ECE 2004

Open solution

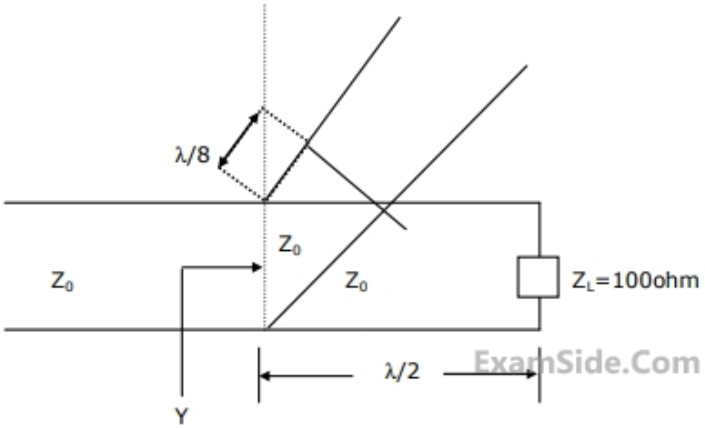
A lossless transmission line is terminated in a load which reflects a part of the incident power. The measured VSWR is 2. The percentage of the power that is reflected back is

(A) 57.73
(B) 33.33
(C) 0.11
(D) 11.11

Q52. GATE ECE 2003

Open solution

A short-circuited stub is shunt connected to a transmission line as shown in Fig. If $Z_0 = 50\ \Omega$, the admittance Y seen at the junction of the stub and the transmission line is



(A) (0.01 − j 0.02) mho
(B) (0.02 − j 0.01) mho
(C) (0.04 + j 0.02) mho
(D) (0.02 + j 0) mho

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Q53. GATE ECE 1999

In a twin-wire transmission line in air, the adjacent voltage maxima are at 12.5 cm and 27.5. The operating frequency is

(A) 300 MHz (B) 1 GHz (C) 2 GHz (D) 6.28 GHz

Open solution

Q54. GATE ECE 1999

In air, a lossless transmission line of length 50 cm with $L = 10\,\mu H/m,\ C = 40\ pF/m$ is operated at 25 MHz. Its electrical path length is

(A) 0.5 meters (B) λ meters (C) $\pi/2$ radians (D) 180 degree

Open solution

Q55. GATE ECE 1994

If a pure resistance load, when connected to a lossless 75-ohm line, produces a VSWR of 3 on the line, then the load impedance can only be 25 ohms.

(A) TRUE (B) FALSE

Open solution

Q56. GATE ECE 1993

Consider a transmission line of characteristic impedance 50 ohm. Let it be terminated at one end by (+ j50) ohm. The VSWR produced by it in the transmission line will be

(A) + 1 (B) 0 (C) ∞ (D) + j

Open solution

Q57. GATE ECE 1992

A transmission line whose characteristic impedance is a pure resistance

(A) must be a lossless line (B) must be a distortionless line

(C) may not be a lossless line (D) may to be a distortionless line

Open solution

Q58. GATE ECE 1991

The input impedance of a short-circuited lossless transmission line quarter wave long is

(A) purely reactive

(B) purely resistive

(C) infinite

(D) dependent on the characteristic impedance of the line

Open solution

Q59. GATE ECE 1989

A 50 ohm lossless transmission line has a pure reactance of (j100) ohms as its load. The VSWR in the line is :

(A) 1/2 (B) 2 (C) 4 (D) ∞

Open solution

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Q60. GATE ECE 1988

A two - wire transmission line of characteristic impedance Z_0 is connected to a load of impedance $Z_L (Z_L \neq Z_0)$. Impedance matching cannot be achieved with

(A) a quarter-wavelength transformer
(B) a half-wavelength transformer
(C) an open-circuited parallel stub
(D) a short-circuited parallel stub

Open solution

Q61. GATE ECE 1987

A transmission line of pure resistive characteristic impedance is terminated with an unknown load. The measured value of VSWR on the line is equal to 2 and a voltage minimum point is found to be at the load. The load impedance is then

(A) complex
(B) Purely capacitive
(C) Purely resistive
(D) Purely inductive

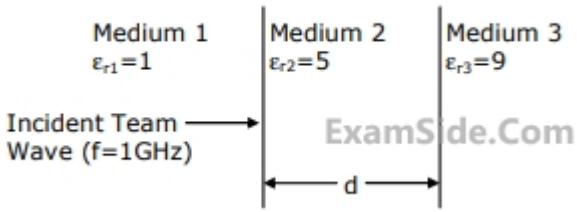
Open solution

Marks 5 2 questions

Q62. GATE ECE 2000

The three regions shown in Fig. are all lossless and non-magnetic. Find
(a) Wave impedance in mediums 2 and 3.
(b) d such that medium 2 acts as a quarter wave $(\lambda/40)$transformer.
(c) Reflection coefficient (Γ) and voltage standing wave ratio (VSWR) at the interface of the medium 1 and 2, when $d = \lambda/4$.

Open solution

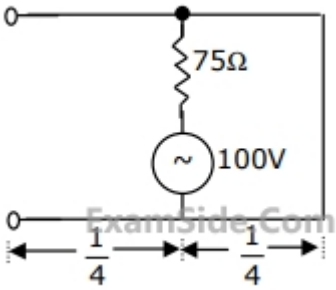


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Q63. GATE ECE 1997

A $\frac{\lambda}{2}$ section of a 600 Ω transmission line, short ciruited at one end and open circuited at the other end, is shown in Fig. A 100 V / 75 Ω generator is connected at the mid point of the section as shown in the Fig. Find the voltage at the open-ciruited end of the line.

Open solution



Marks 10 1 questions

Q64. GATE ECE 1990

A 200 volt (r. m. s) generator having an internal resistance of 200 ohm is feeding a loss-less transmission line. The characteristic impedance and the length of the line are 200 ohm and 10 m respectively. The line is terminated in a resistive load of 100 ohm. If the operating frequency is 37.5 MHz calculate the

Current drawn from the generator.

Magnitude and phase of the current flowing in the load.

Power delivered to the load.

Open solution

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Answer Key

Marks 1

1. **20 to 21**	2. **A**	3. **D**	4. **6.25**	5. **A**
6. **50**	7. **0.87**	8. **B**	9. **B**	10. **70.7**
11. **D**	12. **A**	13. **D**	14. **C**	15. **D**
16. **C**	17. **B**	18. **D**	19. **B**	20. **C**
21. **B**	22. **B**	23. **B**	24. **B**	25. **A**
26. **B**	27. **100 Ω**			

Marks 2

28. **B**	29. **6.3 to 6.5**	30. **200**	31. **B, C**	32. **B, C, D**
33. **0.1**	34. **B**	35. **D**	36. **30**	37. **B**
38. **33.3**	39. **A**	40. **A**	41. **B**	42. **D**
43. **D**	44. **C**	45. **D**	46. **D**	47. **C**
48. **A**	49. **A**	50. **A**	51. **D**	52. **A**
53. **B**	54. **C**	55. **B**	56. **C**	57. **B, C**
58. **C**	59. **D**	60. **B**	61. **C**	

Marks 5

62. **a) Wave impedance in medium** $2 = (60\,\pi)\,\Omega$ **Wave impedance in medium** $3 = (40\,\pi)\,\Omega$ **b) d = 3.75 cm** **c) Reflection coefficient = $-\frac{1}{7}$** **VSWR = 1.33**	63. $800 \angle 90^\circ$ **V**

Marks 10

64. **a)** $\frac{1}{3}\,A$ **b)** $Y_3\ and\ -90^\circ$ **c) 9.875 W**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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